

Short-term residential load forecasting at the portfolio level for a Spanish energy cooperative

Abstract

We document a portfolio-level short-term load forecasting pipeline for a Spanish energy cooperative covering 16,764 residential households. The forecaster is a recursive single-step LightGBM with five quantiles, trained with early stopping on a temporal hold-out followed by a refit on the full 365-day window using the iteration count selected by validation (*refit-on-full*). Split conformal post-hoc calibration brings empirical coverage of the 50% and 90% prediction intervals within 2 percentage points of nominal across three horizon buckets. Walk-forward validation over 105 weekly folds yields a median MAPE of 4.88% at $h \leq 24$ and a 29.6% lift over the persistence baseline at the same horizon. The model underperforms persistence at $h \geq 72$ (lift -25% to -32%), a structural consequence of recursive error compounding that we document explicitly. We also contribute a complementary behavioral segmentation toolkit ($k = 3$, silhouette < 0.20) for tariff design applications, kept architecturally independent from the forecasting pipeline.

1 Introduction

Short-term load forecasting drives day-ahead procurement, intraday balancing and demand-response targeting in retail energy supply. For a small or mid-sized energy cooperative serving residential customers, forecasting at the *portfolio* level (the sum of all customers' hourly consumption) is the operationally relevant target: bulk energy is procured against the aggregate, and imbalance settlement is computed against it. Forecasting *per customer* is unnecessary for procurement and infeasible at scale on residential meters; segmentation *between* those two extremes is a question of architecture.

The hierarchical-forecasting literature (Wickramasuriya et al., 2019; Hyndman and Athanassopoulos, 2021) suggests that forecasting subgroups (e.g. behavioral clusters of customers) and reconciling their forecasts to the portfolio aggregate can improve accuracy when subgroup signals carry distinct information not captured by the aggregate. The trade-off is increased complexity, more models, and a reconciliation step (BottomUp, MinT-OLS, or related trace-minimization estimators) whose benefits depend on whether subgroup heterogeneity is forecastable and stable across the validation window.

This work documents a pipeline that explicitly evaluates the hierarchical-vs-portfolio trade-off and converges on a portfolio-only architecture, with the residual segmentation contribution kept as an independent commercial-side artefact rather than a forecasting input. We report walk-forward MAPE, post-hoc conformal calibration of prediction intervals, and lift over a strong persistence baseline. We document the horizons at which the model wins and at which it loses, the latter being a structural consequence of recursive single-step inference rather than a hyperparameter pathology.

The contribution is methodological rather than novel-architectural: an honest, reproducible pipeline with explicit limitations that other small operators with the same constraints (residential

portfolio, hourly meters, limited tooling budget) can adopt or critique. A complementary *Pipeline 2*, built on the same dataset but architecturally independent, returns a three-archetype k-means segmentation usable for tariff design and demand response targeting.

2 Data

We use the GoiEner smart-meter dataset (Quesada-Granja et al., 2024), a publicly released hourly consumption panel of 25,559 supply points in the Basque Country and other Spanish regions, of which 16,764 are residential (CNAE-2009 code starting with 98). The dataset spans 2014-11-01 through 2022-06-30 with hourly resolution.

After data-quality filtering (minimum one year of valid hours per household, maximum 20% imputed observations, non-zero daily variance), 10,531 households retain a usable pre-2020 daily profile. The remaining 6,233 are dropped: 6,219 lack any pre-2020 observations and 14 have constant readings across the day. Households included after filtering are aggregated to a hourly portfolio panel by summation, yielding 21,671 hourly observations between 2018-01-01 and 2020-06-21 used for champion training, plus the 2020-06-22 to 2022-06-30 validation window over which the walk-forward is computed.

Hourly weather covariates (temperature, relative humidity, precipitation, wind speed) are pulled from the Open-Meteo historical archive at three geographic points (Bilbao, Pamplona, Madrid) and weighted by their share of the cooperative’s customer base (72.5%, 13.8%, 13.7% respectively) to produce a single portfolio-weighted hourly series. Spanish national holidays are obtained from the `holidays` package and matched to calendar dates.

The validation window is intentionally chosen to start in mid-2020 (after the COVID-19 lockdown shock that ended on 2020-06-21) and to span the 2021-06-01 mandatory tariff reform (the 2.0TD time-of-use roll-out), so the walk-forward straddles a known regime change. We address this regime shift in the conformal calibration step by interleaving calibration and test folds rather than splitting them temporally; details follow.

A central caveat: weather features are *realized* (observed historical) rather than forecast. In production, forecast weather would replace the realized series; this paper evaluates methodology rather than deployed accuracy. The same caveat applies in any retrospective forecasting study using historical weather and is documented prominently because it materially affects external validity.

3 Pipeline 1 — Methodology

3.1 Architecture: portfolio-only

We forecast the hourly portfolio aggregate $y_t = \sum_{i \in \text{customers}} y_{i,t}$ directly. Per-segment forecasting was implemented and evaluated under a previous version of this pipeline; we found that (i) the recursive single-step architecture cannot exploit segment heterogeneity at long horizons because the median forecast is fed back as a lag input and over 168 steps the segment trajectories drift toward the mean; and (ii) the validation pinball loss on extreme quantiles ($q = 0.05, 0.95$) for small segments is dominated by tail noise on a 30-day window, producing premature early stopping at iteration counts that did not extrapolate to the prediction horizon. The hierarchical reconciliation step (MinT-OLS, BottomUp) yielded marginal portfolio-MAPE improvements at $h = 25\text{--}72$ but degraded at $h \leq 24$ and $h = 168$, and was therefore dropped together with the per-segment models. Pipeline 1 in this paper is the post-reframe portfolio-only configuration.

3.2 Champion training: early stopping with refit-on-full

For each quantile $q \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$ we train one LightGBM (Ke et al., 2017) booster with quantile loss and recursive single-step target $y_{t+1} \mid \text{features}(t)$. The target window is strictly past at every fold (Section 3.4); we report training in two phases.

Phase 1 runs early stopping on a temporal hold-out. The supervised frame at each training cutoff is split chronologically into 335 days of training and 30 trailing days of validation, with the validation strictly after the training set. `lgb.train` runs with `patience = 100` boosting rounds and a ceiling of 2,500 iterations; the call returns the early-stopping best iteration K . Patience was raised from the LightGBM default (50) after the previous version’s walk-forward exhibited 8.9% of fits stopping under 100 iterations on quantile losses dominated by validation-window noise; raising patience to 100 reduces the rate but does not eliminate it (Section 5).

Phase 2 retrains a fresh booster on the entire 365-day window (training plus the 30-day validation hold-out) for exactly K boosting rounds, with no validation set and no callbacks. This recovers the 30 days sacrificed to early stopping while preserving the iteration count selected by honest validation. The phase-2 booster is the model artefact saved and used downstream. The information that determined K is discarded after phase 1 by construction.

The split is strictly temporal at every step. We assert $\text{valid}_{\min} > \text{train}_{\max}$ before training and abort if the invariant is violated. We verified empirically that across all 105 walk-forward folds the phase-2 maximum timestamp is at least 2 hours before the fold’s issue time, so phase 2 cannot use any data posterior to the cutoff.

LightGBM hyperparameters are: `learning_rate = 0.05`, `num_leaves = 63`, `min_data_in_leaf = 50`, `feature_fraction = 0.9`, `bagging_fraction = 0.9`, `bagging_freq = 5`, fixed across quantiles. Iteration counts of the five champion models on the 2020-06-22 cutoff are $\{784, 642, 510, 184, 305\}$ for $q = \{0.05, 0.25, 0.50, 0.75, 0.95\}$ respectively (`output/05_lgb_training_summary.csv`). The asymmetric distribution (extreme quantiles taking more iterations than the upper-mid $q = 0.75$) reflects the differing convergence rates of the asymmetric pinball losses.

3.3 Walk-forward validation and conformal calibration

Walk-forward generates 105 issue points spaced 7 days apart across 2020-06-22 to 2022-06-20, each issuing a 168-hour forecast trajectory. At every issue point we refit the five-quantile suite on the prior 365 days following the phase-1 plus phase-2 protocol of Section 3.2. Three folds at the dataset tail are skipped because the 168-hour target window falls outside the available data, leaving 102 evaluable folds and $102 \times 5 = 510$ booster fits in total.

Recursive forecasts are produced one step at a time: at step 1, features at the issue time predict $\hat{y}_1$; at step 2, the median forecast is fed into the lag 1 feature slot to predict $\hat{y}_2$; and so on through step 168. Lag-24 and lag-168 features are reconstructed at every step from the working trajectory’s earlier predictions, while calendar and weather features are read directly from the deterministic exogenous block.

Conformal calibration follows split conformal prediction (Vovk et al., 2005; Shafer and Vovk, 2008). The 102 issue points are sorted in time and split into calibration and test sets by index parity (even indices $\rightarrow$ calibration, odd $\rightarrow$ test): with the validation period straddling the 2021-06-01 tariff reform, a naive temporal split would put pre-reform issues entirely in calibration and post-reform entirely in test, breaking exchangeability. Index parity distributes the regime shift evenly across the two halves.

For each pool bucket $b \in \{h \leq 24, h = 25\text{--}72, h = 73\text{--}168\}$ and each non-median quantile $q \in \{0.05, 0.25, 0.75, 0.95\}$, we estimate

$$\delta(b, q) = \text{Quantile}_q(y_{\text{actual}} - \hat{y}_q) \quad \text{on calibration,}$$

and apply $\hat{y}_q^{\text{cal}} = \hat{y}_q + \delta(b, q)$ at test time. The median ($q = 0.5$) is left unchanged, so median MAPE is preserved by construction. Pooling across horizons within a bucket lifts the calibration sample size from ~ 51 per horizon to 1,224 / 2,448 / 4,872 per (bucket, quantile), at the cost of losing per-horizon resolution; the README headline coverage figures are reported at the bucket level for this reason.

3.4 Features and causality

The feature block is intentionally minimal. Lag features $y_{t-1}, y_{t-24}, y_{t-168}$ encode hour-of-day, day-of-week and weekly seasonality. Past-only rolling means $\text{mean}(y_{t-24}, \dots, y_{t-1})$ and $\text{mean}(y_{t-168}, \dots, y_{t-1})$ are computed via `pandas.shift(1).rolling(window=N).mean()`, which excludes y_t and makes the rolling value at t a strict function of past observations. Calendar features (hour, day-of-week, weekend, month, day-of-year, sine and cosine encodings, Spanish holiday indicator) are deterministic functions of the timestamp. Weather features (`temperature_weighted`, `humidity_weighted`, `precipitation_weighted`, `wind_weighted`, plus row-local nonlinear transforms `temp_sq`, `heating_demand`, `cooling_demand`) are realized at row time. A single regime indicator `post_tariff_reform` flags timestamps from 2021-06-01 onward.

Every feature is row-local: at row t the feature value depends only on inputs at t or at $t - N$ for fixed $N \geq 1$. There is no exponentially-weighted mean, no centered rolling window, no normalization with global statistics, and no encoding based on global frequencies. The target y_{t+1} uses positive shift, with the supervised filter requiring $t < \text{train_end} - 1\text{h}$ so that the target itself stays strictly inside the fold window. This causality property was checked empirically across the 105 folds before training (Section 3.2).

4 Pipeline 2 — Methodology

Pipeline 2 is a behavioural segmentation toolkit, kept architecturally independent from Pipeline 1. It does not feed features into the forecasting model and is not consumed by the daily batch run. We document it because the same operator that runs the forecasting pipeline often needs a coarse, interpretable customer segmentation for tariff design, demand-response targeting, or commercial reporting.

For each household with at least one full year of pre-2020 hourly data we compute the 24-hour mean profile $\bar{p}_{i,h}$ ($h = 0, \dots, 23$), standardize it per-household to remove level ($\hat{p}_{i,h} = (\bar{p}_{i,h} - \mu_i) / \sigma_i$), and run k-means on the resulting $10\,531 \times 24$ matrix. The pre-2020 cutoff matches the validation start so the segmentation does not see post-pandemic or post-tariff-reform regime shifts.

We choose k by sweeping $\{2, 3, 4, 5, 6, 7, 8\}$ and reporting the silhouette score (sub-sample $n = 2000$, fixed seed), the Davies-Bouldin index, the Calinski-Harabasz index and the within-cluster inertia. Across all k , silhouette stays below 0.20: residential daily load shape is a continuum, not a small set of natural clusters. The metrics disagree on the optimum (silhouette peaks at $k = 3$, Davies-Bouldin minimizes at $k = 6$, Calinski-Harabasz at $k = 2$). The elbow plot inflects most sharply between $k = 2$ and $k = 3$ (inertia drop $-13\,003$, against $-9\,197$ from $k = 3$ to $k = 4$). We report $k = 3$ as the metric-supported and operationally interpretable choice.

The three resulting archetypes are summarised in Table 1. We tag each centroid with an operational label using a two-tier heuristic: when the centroid amplitude exceeds 2σ in the standardized profile, we tag it by the dominant 5-period mean (night, morning, midday, afternoon, evening);

Table 1: Pipeline 2 archetypes (k-means with $k = 3$ on standardized pre-2020 daily profiles, 10,531 households).

seg	label	households	% hh	% kWh	mean kWh/hh	peak h	amp (σ)
0	evening-peak	5,624	53.4%	46.4%	3,966	22:00	3.14
1	high-baseload	729	6.9%	12.3%	8,128	00:00	1.82
2	midday-active	4,178	39.7%	41.3%	4,747	11:00	2.12

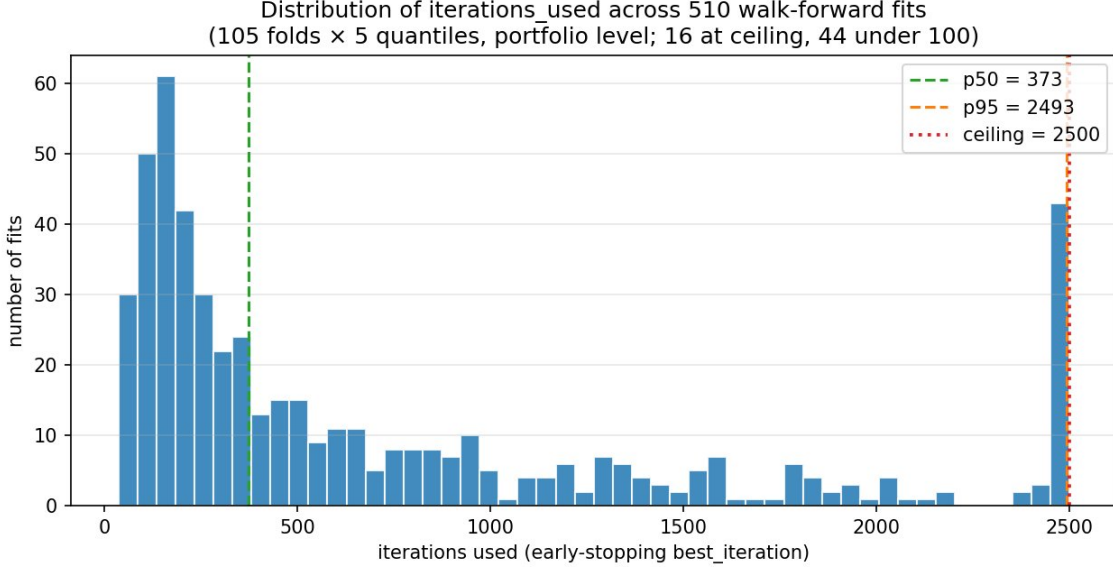


Figure 1: Distribution of `iterations_used` (LightGBM `best_iteration`) across 510 walk-forward fits, with the 50th and 95th percentiles and the 2,500-round ceiling marked. 16 fits (3.1%) hit the ceiling; 44 (8.6%) stop under 100 iterations.

when the amplitude is below 2σ the daily swing is not informative and we tag by the level relative to the portfolio mean (**high-baseload** for $> 1.5\times$, **low-flat** for $< 0.7\times$, **flat-medium** otherwise). The labels are operational tags, not statistical discoveries; the silhouette < 0.20 caveat is prominent everywhere these archetypes are reported.

5 Results

5.1 Iteration distribution

Figure 1 shows the distribution of LightGBM `best_iteration` across the 510 walk-forward fits. The median is 375 iterations, the 90th percentile is 2,152, and 16 fits (3.1%) hit the ceiling of 2,500 boosting rounds. At the other end, 44 fits (8.6%) stop in fewer than 100 iterations: these are concentrated in the quantile-tail fits ($q = 0.05$ contributes the largest share) where the 30-day pinball-loss validation signal is dominated by a handful of tail observations and the patience criterion exits on noise. Raising patience from 50 (the default) to 100 reduced this rate but does not eliminate it, consistent with our diagnosis that the residual cause is irreducible validation-window noise rather than insufficient patience.

Table 2: Walk-forward median MAPE by horizon, portfolio level, n = total target rows over 102 evaluable folds.

horizon	n	persistence	SARIMAX	LightGBM
$h \leq 24$	2,448	6.93%	13.38%	4.88%
$h = 25\text{--}72$	4,896	6.29%	17.44%	5.20%
$h = 73\text{--}168$	9,768	7.11%	23.11%	6.26%
$h = 1$	102	5.23%	16.83%	3.58%
$h = 24$	102	4.97%	12.09%	4.02%
$h = 72$	102	4.37%	13.69%	5.46%
$h = 168$	101	5.25%	16.79%	6.95%

Table 3: Empirical coverage on the test split (51 issue points), raw recursive bands vs. conformal-calibrated bands. The 50% and 90% nominal intervals correspond to $[q_{0.25}, q_{0.75}]$ and $[q_{0.05}, q_{0.95}]$ respectively.

bucket	n	raw cov_50	cal cov_50	raw cov_90	cal cov_90
$h \leq 24$	1,224	0.219	0.475	0.546	0.899
$h = 25\text{--}72$	2,448	0.184	0.490	0.475	0.878
$h = 73\text{--}168$	4,872	0.177	0.482	0.463	0.883
$h = 1$	51	0.451	0.784	0.686	0.922
$h = 24$	51	0.118	0.471	0.373	0.882
$h = 72$	51	0.020	0.471	0.314	0.824
$h = 168$	50	0.120	0.380	0.380	0.820

5.2 Forecast accuracy by horizon

Table 2 reports walk-forward median MAPE per horizon bucket and per pointwise horizon for the three competing forecasters: a weekly persistence baseline ($\hat{y}_t = y_{t-168}$), a SARIMAX baseline with exogenous block, and the LightGBM portfolio-only suite. The headline result is MAPE = 4.88% at $h \leq 24$, ahead of persistence (6.93%) and SARIMAX (13.38%) by margins of 2.05 and 8.50 percentage points respectively.

The LightGBM model wins across all three pool buckets, but the pointwise breakdown reveals that the win at $h \leq 24$ is decisive (3.58% at $h = 1$ vs. 5.23% for persistence) while at $h = 72$ and $h = 168$ LightGBM *loses* to persistence (5.46% vs. 4.37% and 6.95% vs. 5.25% respectively). We discuss the mechanism in Section 5.4 and as a limitation in Section 6.

5.3 Coverage post-conformal

Table 3 summarizes empirical coverage of the 50% and 90% prediction intervals on the test split before and after the conformal δ adjustment. The three pool buckets all satisfy the calibration gate ($[0.45, 0.55]$ for cov_50, $[0.85, 0.95]$ for cov_90) post-calibration. Pre-calibration the recursive bands substantially undercover at the median horizons: cov_50 drops to 0.18 at $h = 25\text{--}72$ and to 0.18 at $h = 73\text{--}168$, with cov_90 similarly compressed. The conformal adjustments $\delta(b, q)$ are negative for the lower quantiles and positive for the upper quantiles, widening the bands by an amount that grows with the pool’s distance from $h = 1$ (median δ for $q = 0.05$ moves from -128 at $h \leq 24$ to -216 at $h = 73\text{--}168$).

The pointwise diagnostic rows ($h \in \{1, 24, 72, 168\}$) are reported for transparency but are not

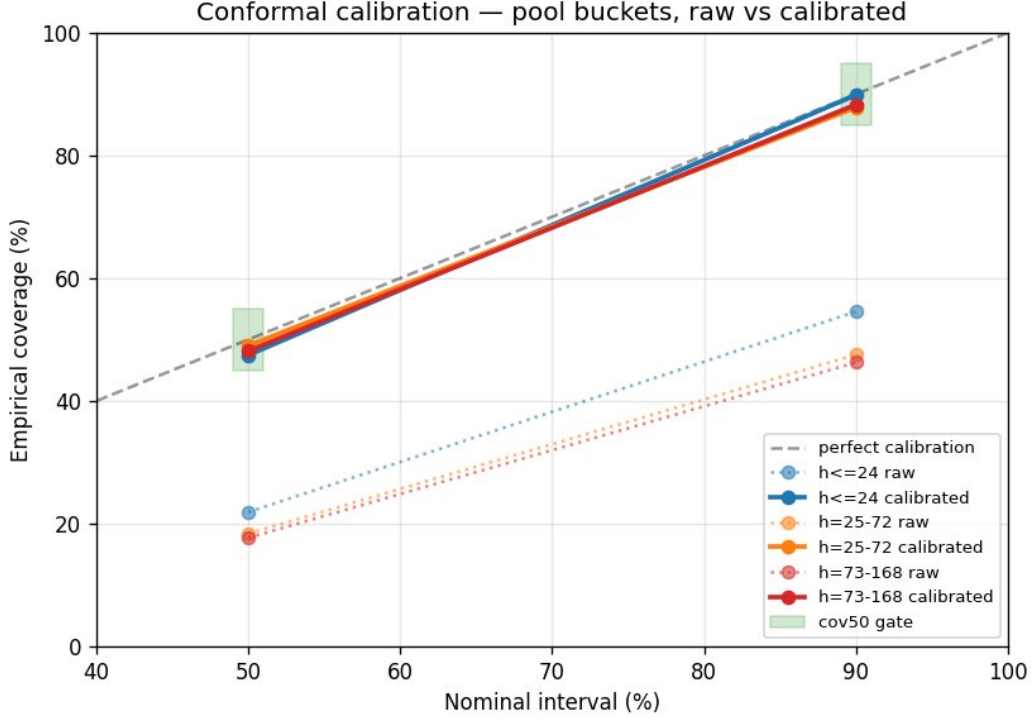


Figure 2: Conformal calibration: empirical coverage versus nominal interval, raw recursive bands (dotted) and conformal-calibrated bands (solid), pooled by horizon bucket. Green strips mark the $[0.45, 0.55] \times [0.85, 0.95]$ acceptance region.

part of the calibration gate: the δ adjustments are computed at the pool-bucket level and within-pool drift is by design. The visible undercoverage at $h = 168$ (cal cov₉₀ = 0.82) is the deepest case of within-pool drift and would require per-horizon adjustments to fix at the cost of much smaller calibration samples.

Figure 2 plots empirical coverage at the 50% and 90% nominal points, raw vs. calibrated, for the three pool buckets, showing the post-calibration trajectories converging close to the diagonal of perfect calibration.

5.4 Lift over persistence

Table 4 reports the relative MAPE improvement of LightGBM over weekly persistence, $\text{lift} = (\text{MAPE}_{\text{persist}} - \text{MAPE}_{\text{LGB}}) / \text{MAPE}_{\text{persist}}$.

LightGBM dominates persistence at horizons short enough that the median recursive prediction has not drifted appreciably from the realized trajectory. At pointwise $h = 72$ and $h = 168$ the situation reverses: the recursive median has been folded back as a lag input dozens or hundreds of times, accumulating per-step error that cannot be undone by the model’s last-step output. Persistence weekly — which simply copies the value from one week earlier — is a remarkably strong baseline at these horizons because residential consumption is highly periodic at the weekly scale. This is a structural property of recursive single-step inference, not a hyperparameter problem; we discuss the implication in Section 6.

Table 4: LightGBM relative MAPE improvement over weekly persistence $\hat{y}_t = y_{t-168}$, by horizon. Negative lift indicates the LightGBM model loses to the baseline.

horizon	persistence MAPE	LightGBM MAPE	lift
$h = 1$	5.23%	3.58%	+31.6%
$h \leq 24$	6.93%	4.88%	+29.6%
$h = 24$	4.97%	4.02%	+19.1%
$h = 25-72$	6.29%	5.20%	+17.3%
$h = 73-168$	7.11%	6.26%	+12.0%
$h = 72$	4.37%	5.46%	-25.0%
$h = 168$	5.25%	6.95%	-32.4%

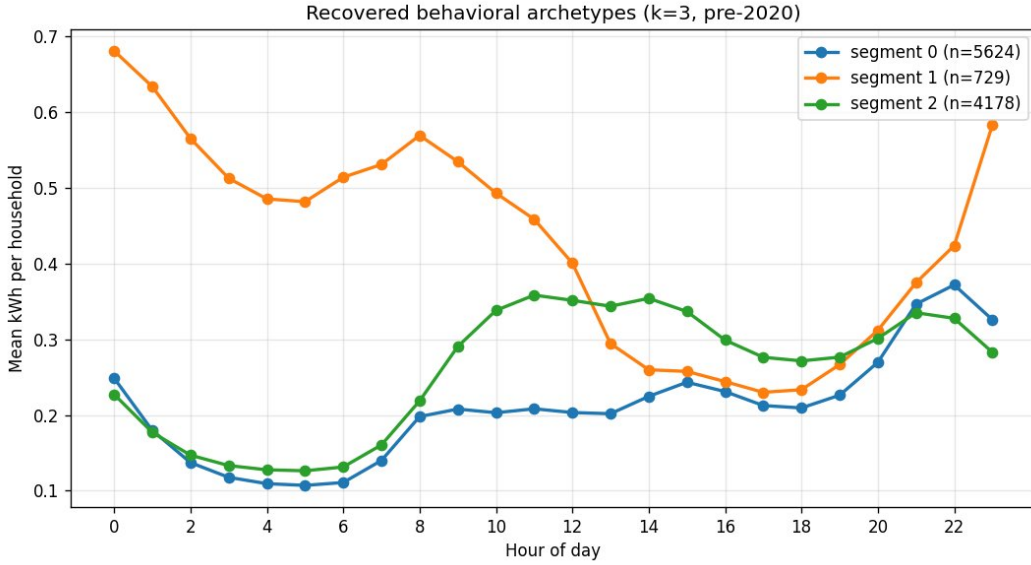


Figure 3: Recovered behavioral archetypes (k-means with $k = 3$, pre-2020). Each curve plots the within-cluster mean kWh per household by hour of day in raw kWh; cluster sizes shown in legend.

5.5 Behavioral archetypes

Figure 3 plots the three Pipeline 2 archetypes in the raw kWh-per-household scale. The **evening-peak** archetype (53.4% of households, 46.4% of kWh) shows the canonical residential pattern of low day-time consumption and a 21:00–23:00 peak. The **high-baseload** archetype (6.9% of households, 12.3% of kWh) sustains 0.5–0.7 kWh/h through the night and is the highest per-customer consumer ($1.78\times$ portfolio mean). The **midday-active** archetype (39.7% of households, 41.3% of kWh) ramps from 09:00 and holds through the afternoon, consistent with retired households, work-from-home, and households with day-time caregiving.

The commercial reading is straightforward: **evening-peak** is the target population for time-of-use tariffs that incentivize off-peak shifting; **high-baseload** is the highest-revenue segment per customer and the natural candidate for retention or premium service positioning; **midday-active** is the segment whose load profile best matches photovoltaic self-consumption and is therefore a natural target for solar-bundle commercial offers. None of these readings depend on the silhouette being high; the $k = 3$ partition is operationally interpretable on the raw kWh scale even when the cluster boundaries are soft, which they are here.

6 Limitations

Recursive error compounding at long horizons. LightGBM underperforms persistence at $h = 72$ and $h = 168$ (lift -25% and -32% respectively). The recursive single-step architecture predicts $\hat{y}_{t+1}$ given features that include past consumption, then folds $\hat{y}_{t+1}$ back into the lag-1 slot to predict $\hat{y}_{t+2}$, and so on through the 168-step horizon. The accumulated error is not unbiased under quantile loss: the per-step residual distribution is fed into a nonlinear ensemble whose forward error grows superlinearly in step count. A direct multi-horizon architecture — one model per horizon, trained on $y_{t+h} \mid \text{features}(t)$ rather than y_{t+1} — would address this at the cost of $7\times$ to $168\times$ more models depending on horizon granularity. We did not implement it; this is the natural next iteration.

Pinball-loss noise on extreme quantiles. 8.6% of fits stop in fewer than 100 iterations, concentrated at $q = 0.05$ and $q = 0.95$ where the validation pinball loss over a 30-day window is dominated by a handful of tail observations. Patience = 100 reduces but does not eliminate this. Two paths forward, both out of scope here: (i) widen the validation window to 60 or 90 days, accepting that more recent training data is sacrificed, or (ii) replace patience-based early stopping with a proper Bayesian iteration prior calibrated on prior folds. We document the rate so it can be revisited if it materially affects deployed coverage.

Realized weather in features. Weather covariates are observed historical, not forecast. In production, day-ahead and intraday weather forecasts (with their own error distribution) would replace the realized series. The MAPE numbers reported here therefore overstate deployed accuracy by an amount that grows with the contribution of weather features to the model. Feature-importance aggregation across the five champion models attributes 20.5% of total gain to the weather block, so the expected deployment-time degradation is non-trivial.

Iteration ceiling activated. 3.1% of fits hit the 2,500-round ceiling, all concentrated at the high-quantile end of the distribution. This indicates that for some folds the model would have benefited from more iterations than the cap allowed. Raising the cap to 3,500 or 4,000 would clear these but at the cost of run time and an unknown benefit; we kept the cap conservative.

Pipeline 2 silhouette below 0.20. At every k in the sweep, the silhouette stays below the conventional 0.20 threshold for “reasonable structure” (and well below the 0.50 threshold for “strong structure”). Residential daily load shape is a continuum, not a small set of natural clusters. The $k = 3$ partition is reported as an operational interpretable carving, not a statistical discovery. Any commercial use of the labels needs to acknowledge this or risk treating soft, overlapping groups as if they were strong types.

Single-cooperative geographic scope. The dataset covers one Spanish cooperative, predominantly Basque Country residential customers. The portfolio composition (large **evening-peak** share, small **high-baseload** share) is local to that customer base and is not representative of Spanish residential supply at large, let alone other markets. External validity claims should be limited accordingly.

7 Operational implications and conclusion

The lift over persistence at $h \leq 24$ (29.6%) and $h = 25\text{--}72$ (17.3%) is large enough to justify adopting LightGBM at the portfolio level for short-horizon procurement and intraday balancing applications. The deterioration past $h = 72$ is structural; for week-ahead applications the weekly persistence baseline $\hat{y}_t = y_{t-168}$ remains competitive, and we would recommend either using persistence directly at those horizons or moving to a direct multi-horizon architecture before deploying recursive single-

step at $h = 168$.

Conformal calibration brings the 50% and 90% prediction intervals within 2 percentage points of nominal across the three pool buckets, satisfying the calibration gate without modifying the median forecast. Intervals are usable for hedging and reserve sizing under the explicit caveat that they are calibrated at the pool-bucket level rather than at each individual horizon, with the largest within-pool drift at $h = 168$ where `cov_90` drops to 0.82.

Pipeline 2 is delivered as a complementary commercial-side artefact: three operationally interpretable archetypes with explicit silhouette < 0.20 caveat, usable for tariff design and demand-response targeting but not as feature input to the forecasting pipeline. The two pipelines share the data ingestion stage but diverge thereafter, which we view as the correct architectural split given that segmentation for forecasting and segmentation for commercial use have different success criteria.

The contribution is methodological, not algorithmic: an honest pipeline documented with its trade-offs, suitable for adoption by other small operators with similar constraints. Source code, intermediate outputs and this paper are versioned together in the project repository.

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